Gold Standard Science, Communication, and Public Trust

FCSM Research and Policy Conference
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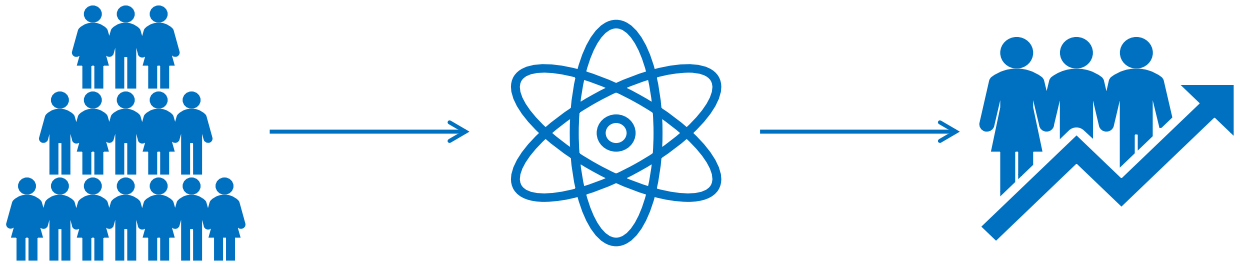
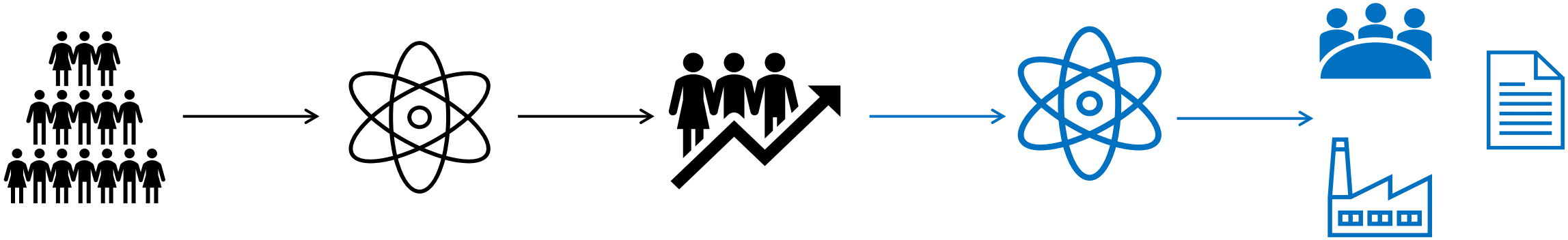
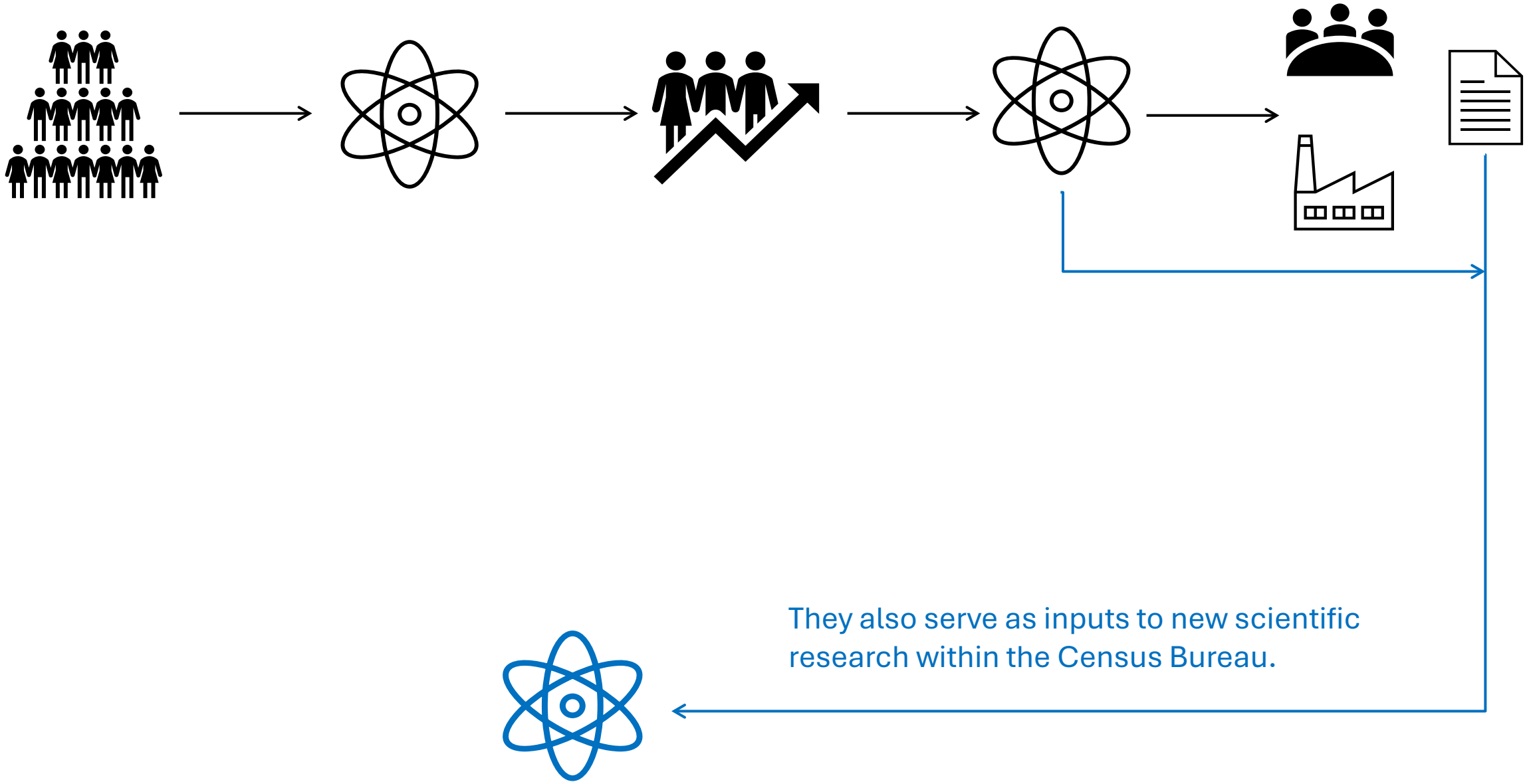
The generation and use of science is central to everything the Census Bureau does.

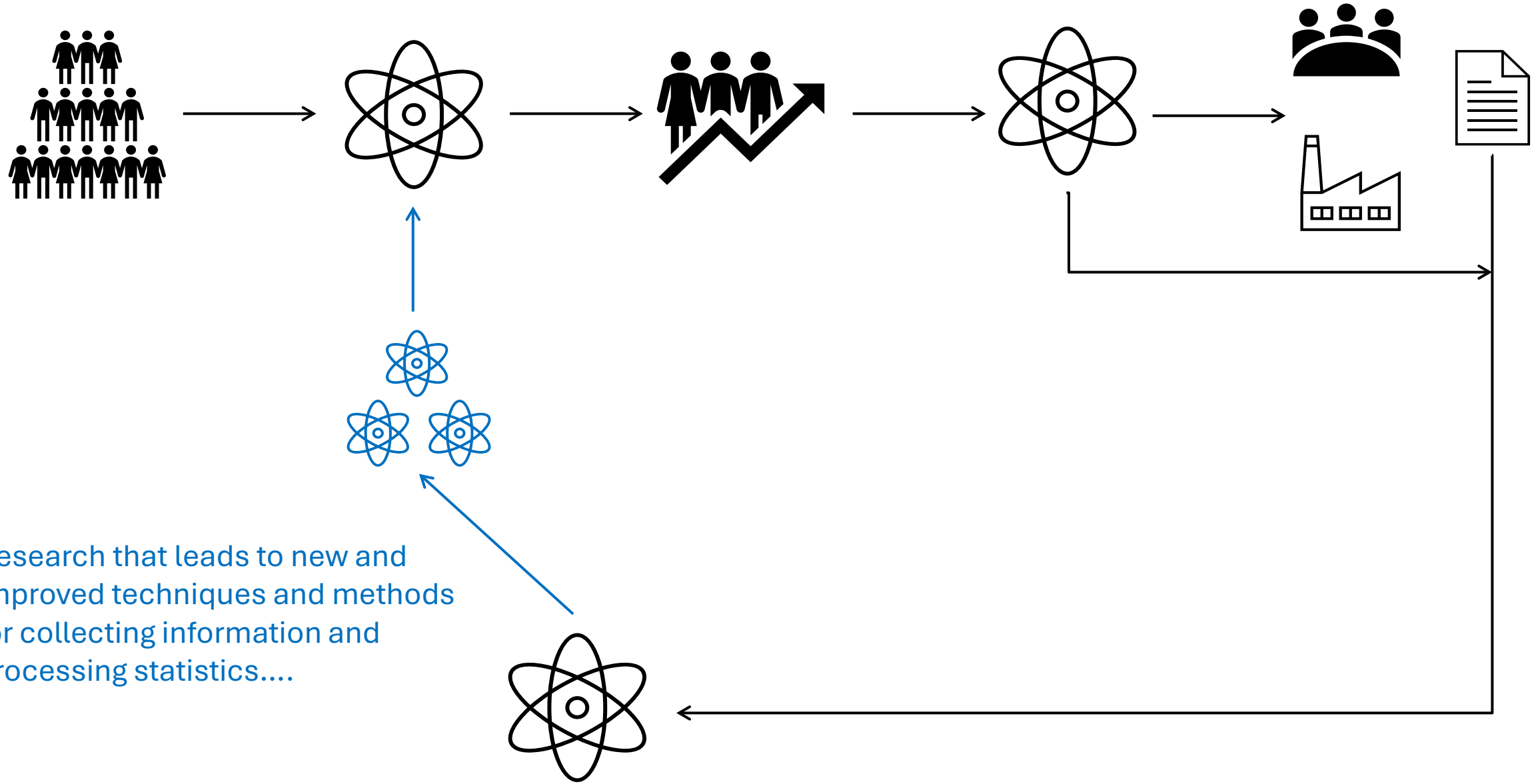


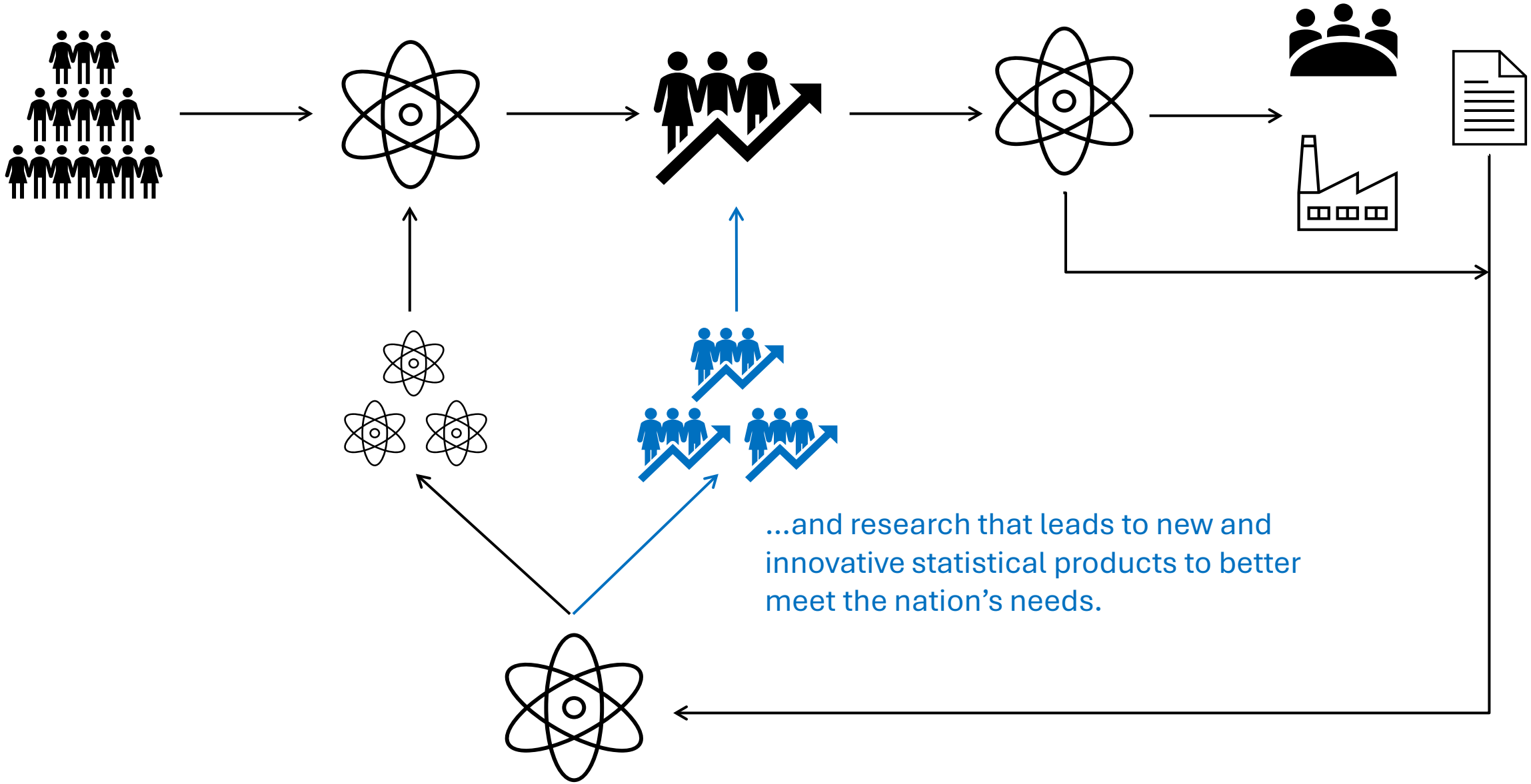
We use scientific methods to collect information about the nation’s people and economy, and to process that information into our various statistical products.

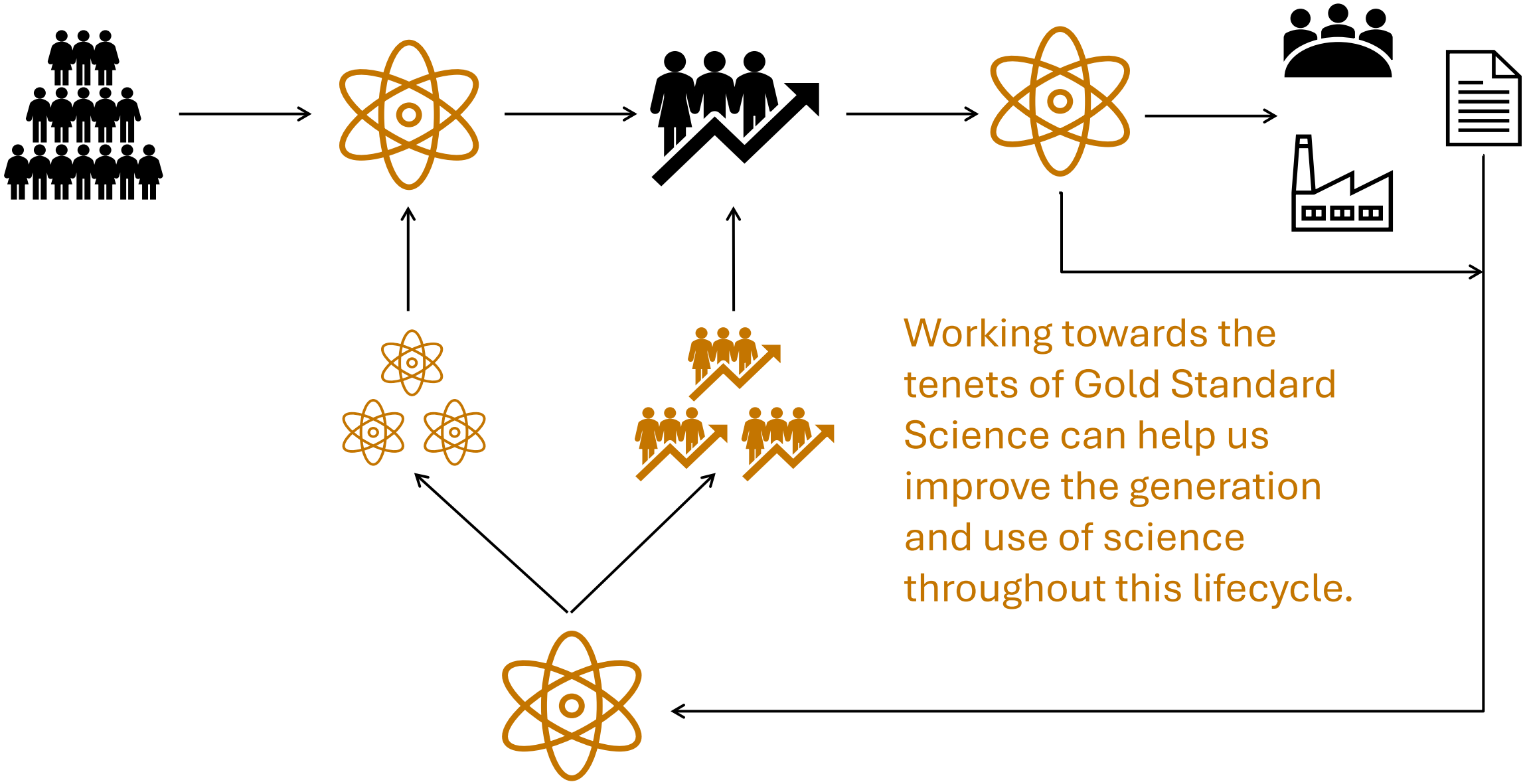


Those statistical products then serve as important scientific inputs to analyses evaluations, and decision-making by policymakers, businesses, and researchers.


They also serve as inputs to new scientific research within the Census Bureau.


Research that leads to new and improved techniques and methods for collecting information and processing statistics....


...and research that leads to new and innovative statistical products to better meet the nation's needs.


Working towards the tenets of Gold Standard Science can help us improve the generation and use of science throughout this lifecycle.

NATIONAL ARCHIVES

FEDERAL REGISTER

The Daily Journal of the United States Government

Presidential Document

Restoring Gold Standard Science

A Presidential Document by the Executive Office of the President on 05/29/2025

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PDF

Document Details

Executive Order Details

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(printed page 22601)

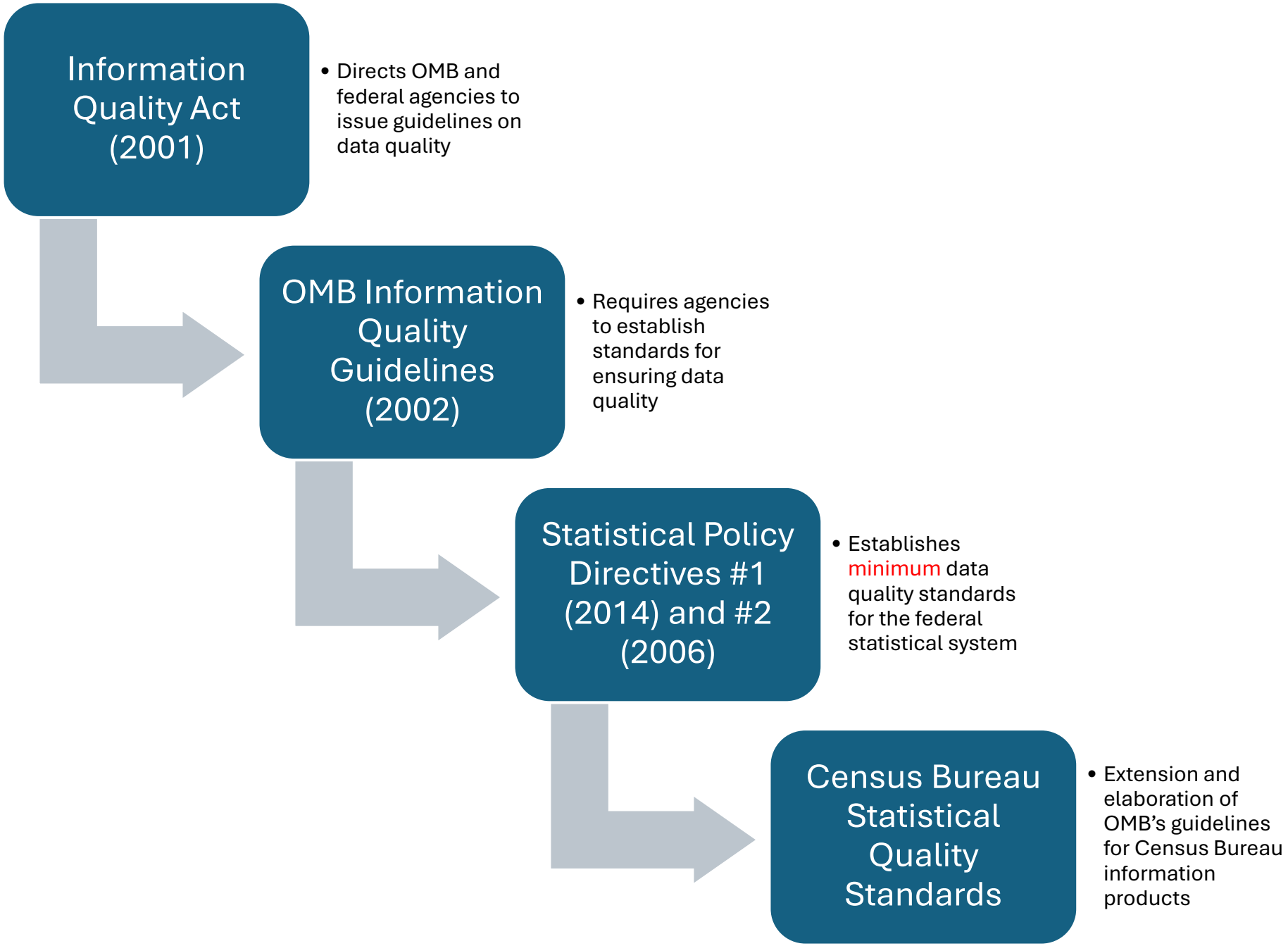
Executive Order 14303 of May 23, 2025

Restoring Gold Standard Science

By the authority vested in me as President by the Constitution and the laws of the United States of America, including section 7301 of title 5, United States Code, it is hereby ordered:

Where does Gold Standard Science fit within existing legal and policy frameworks governing the quality of our statistics?

...and how does it supplement and enhance these frameworks?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OMB Information Quality Guidelines (2002)
• Requires agencies to establish standards for ensuring data quality
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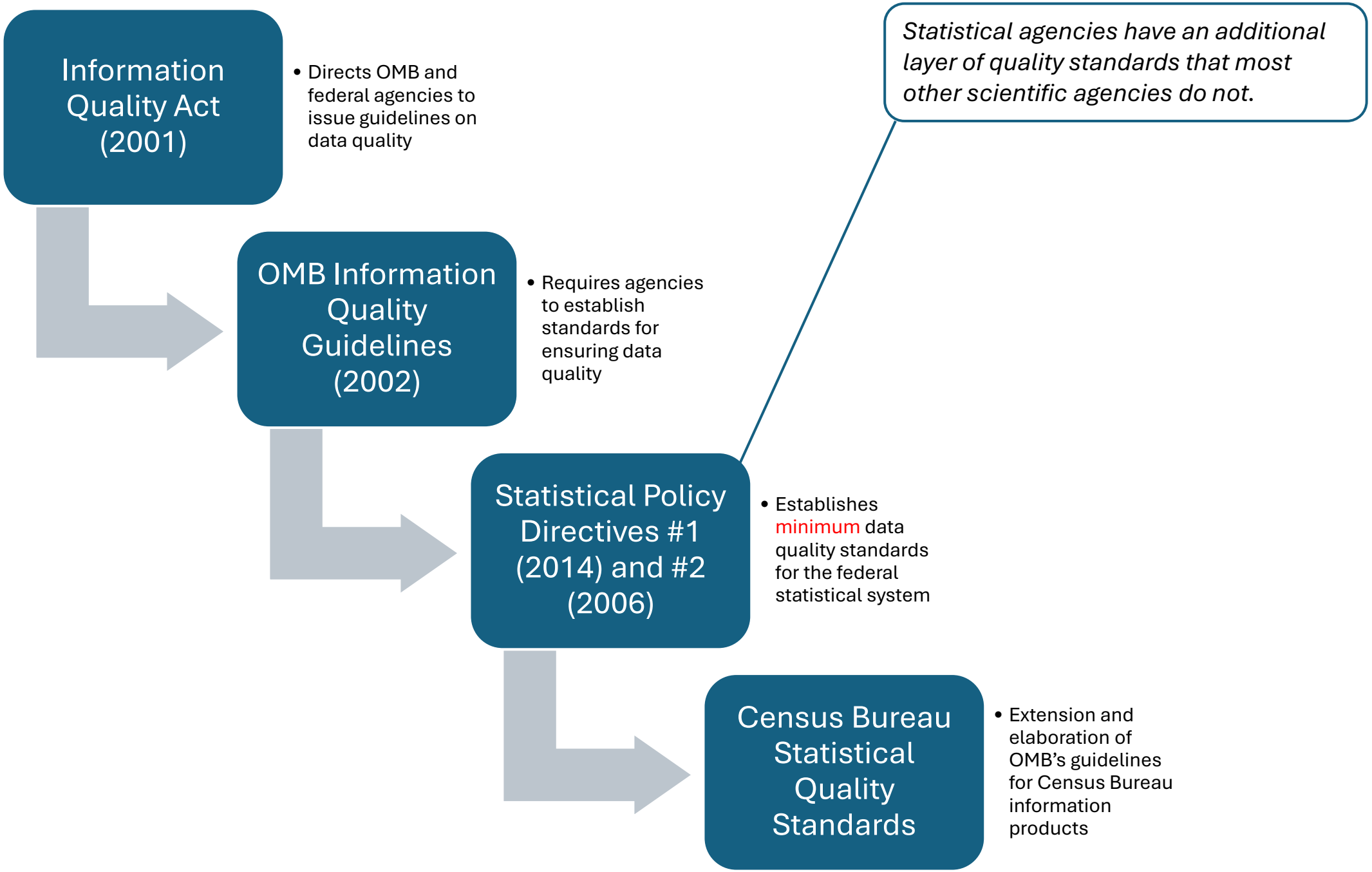
Census Bureau Statistical Quality Standards
• Extension and elaboration of OMB's guidelines for Census Bureau information products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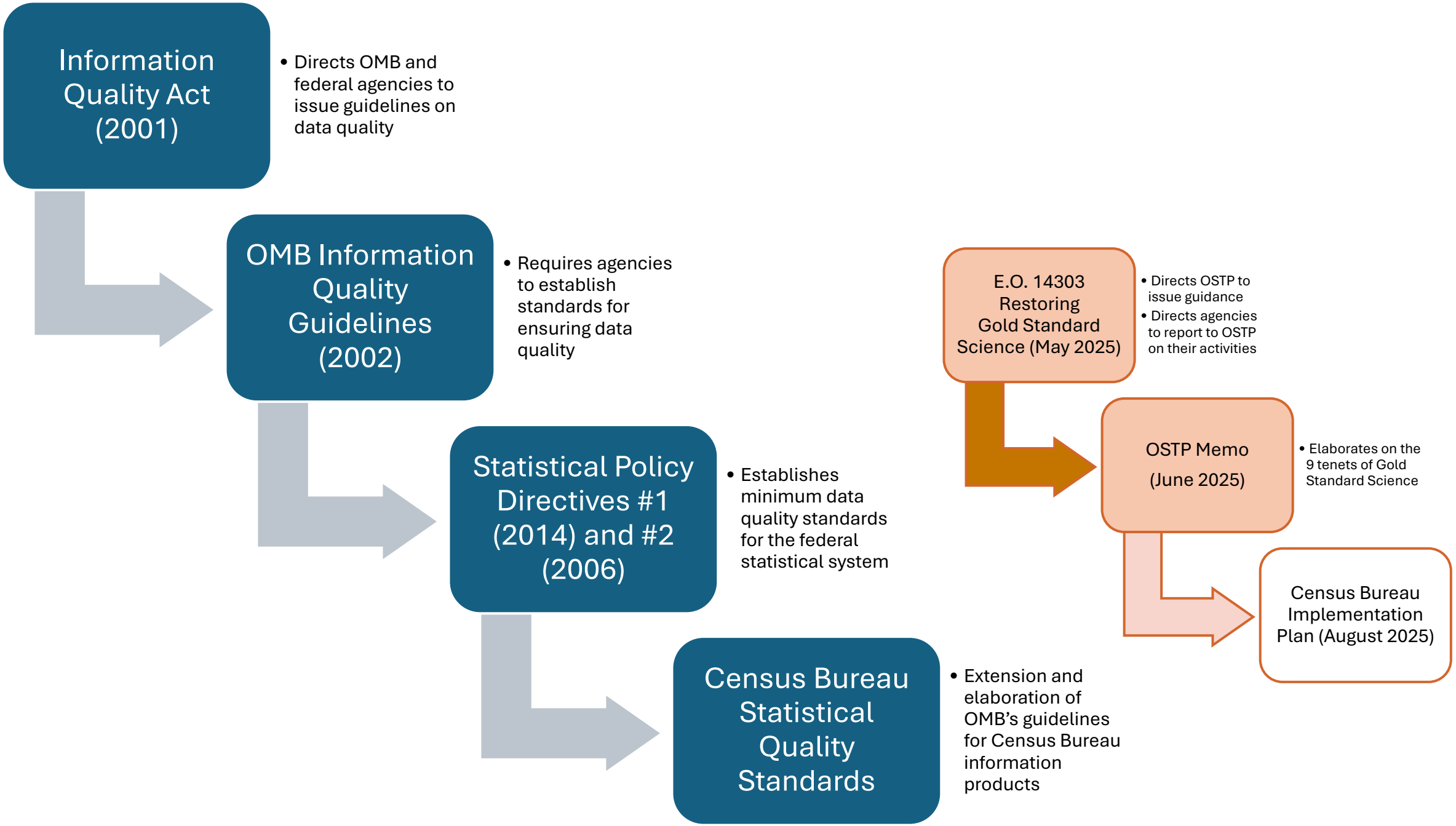
Statistical agencies have an additional layer of quality standards that most other scientific agencies do not.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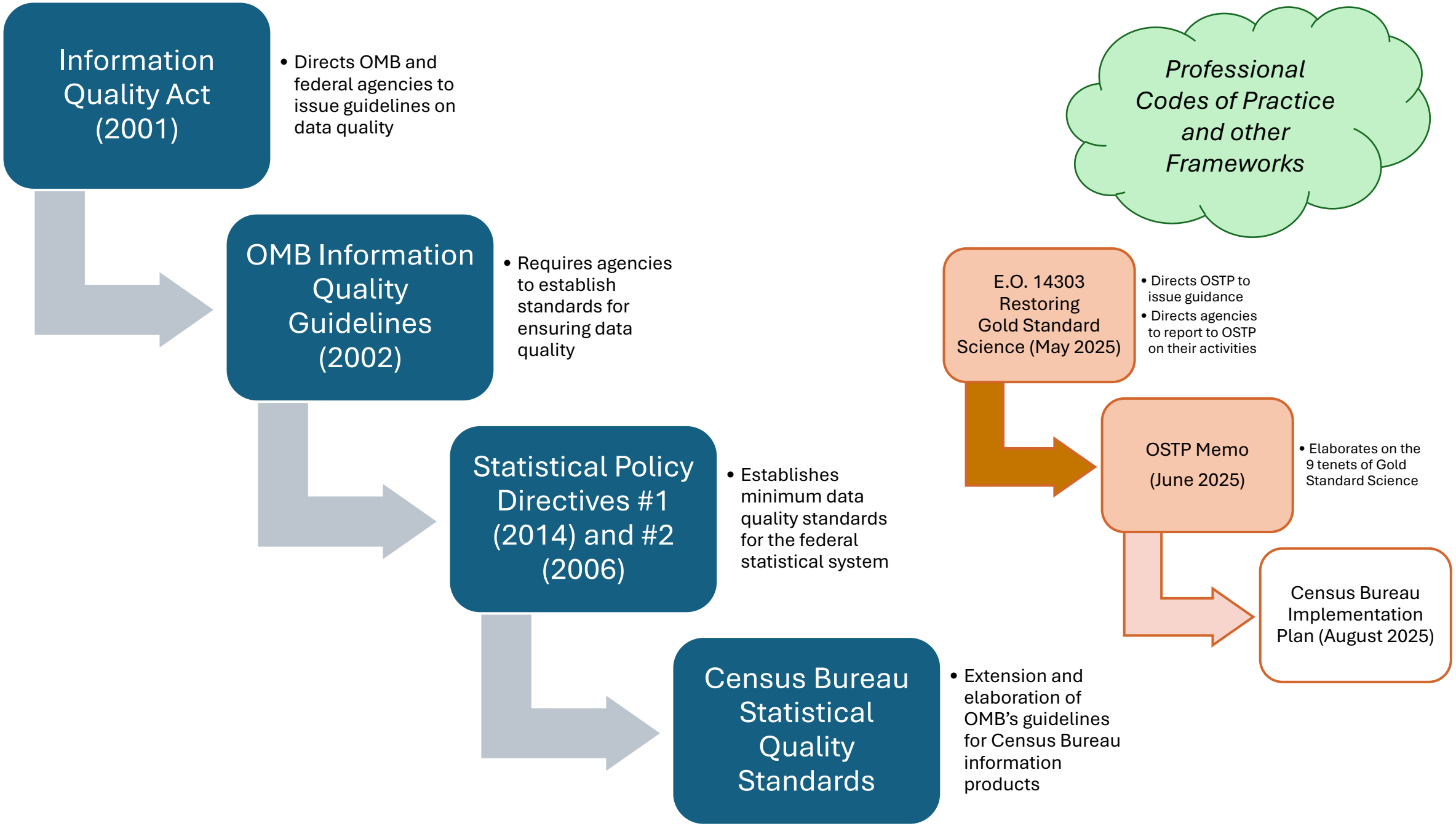
• Directs OSTP to issue guidance
• Directs agencies to report to OSTP on their activities
OSTP Memo (June 2025)
• Elaborates on the 9 tenets of Gold Standard Science
Census Bureau Implementation Plan (August 2025)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Tenets of Gold Standard Science

Reproducible

Accepting of Negative Results as Positive Outcomes

Skeptical of Findings and Assumptions

Communicative of Error and Uncertainty

Structured for Falsifiability of Hypotheses

Without Conflict of Interest

Collaborative and Interdisciplinary

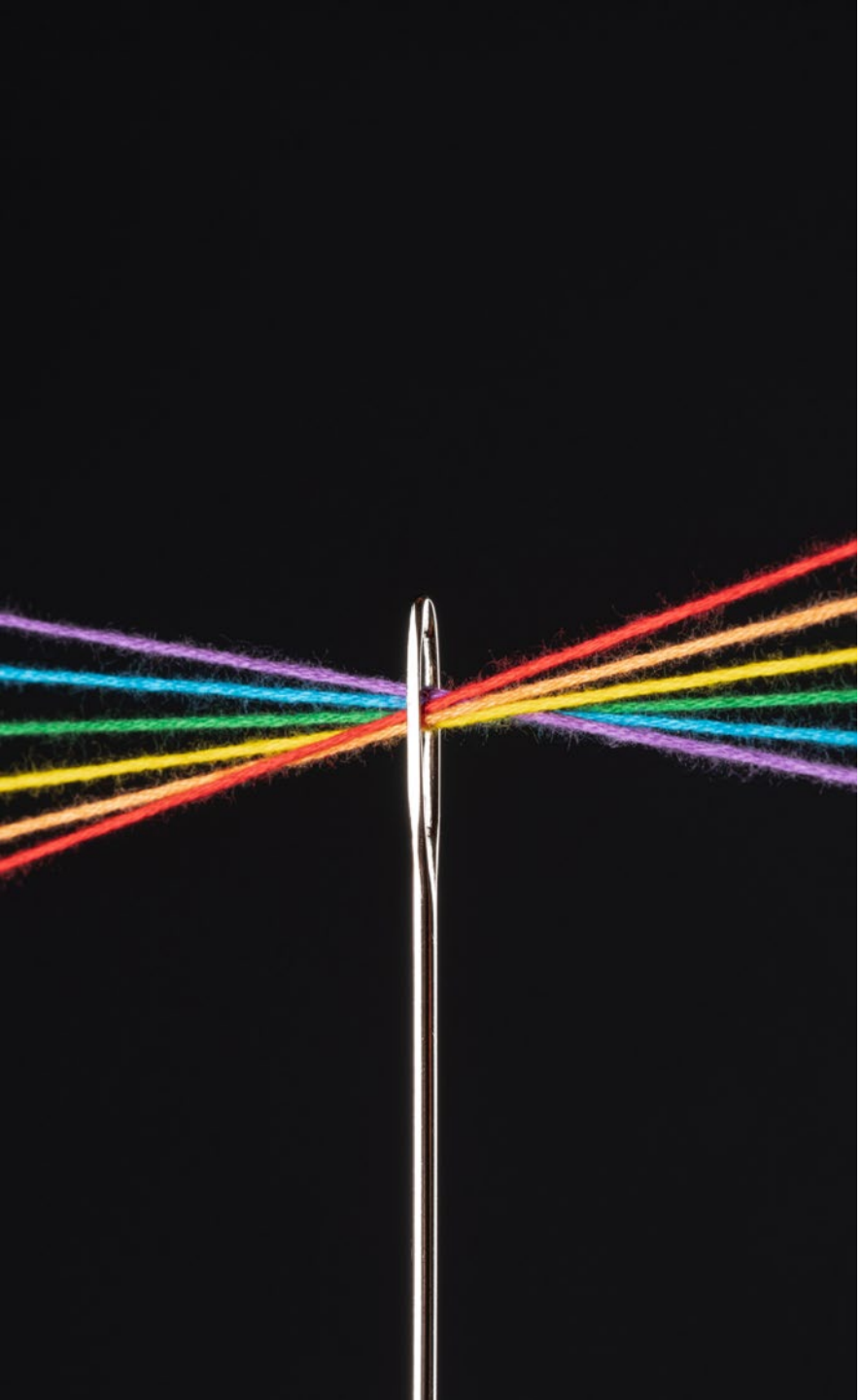
Unbiased Peer Review

Transparent

Scientific Rigor,
Professional Codes of Practice,
Internal & External Peer Review

Operationalizing the Gold Standard Science Tenets

What are we doing already, and what do we need to do?



Collaborative and Interdisciplinary

Already Established

Inclusive publication practices

Formal review structures that cross directorates and disciplines

Strong culture of collaboration

Future Enhancements

Bureau-wide training on the role of science—develop and promote training on the role of science in Bureau activities, stressing the importance of scientific rigor and the responsible conduct of research reflecting differing norms and practices from the different disciplines

Structured for Falsifiability of Hypotheses

Already Established

Pilot Programs

Growing acceptance that not all pilot programs will succeed, and not all research-based and experimental statistical products will transition to core products.

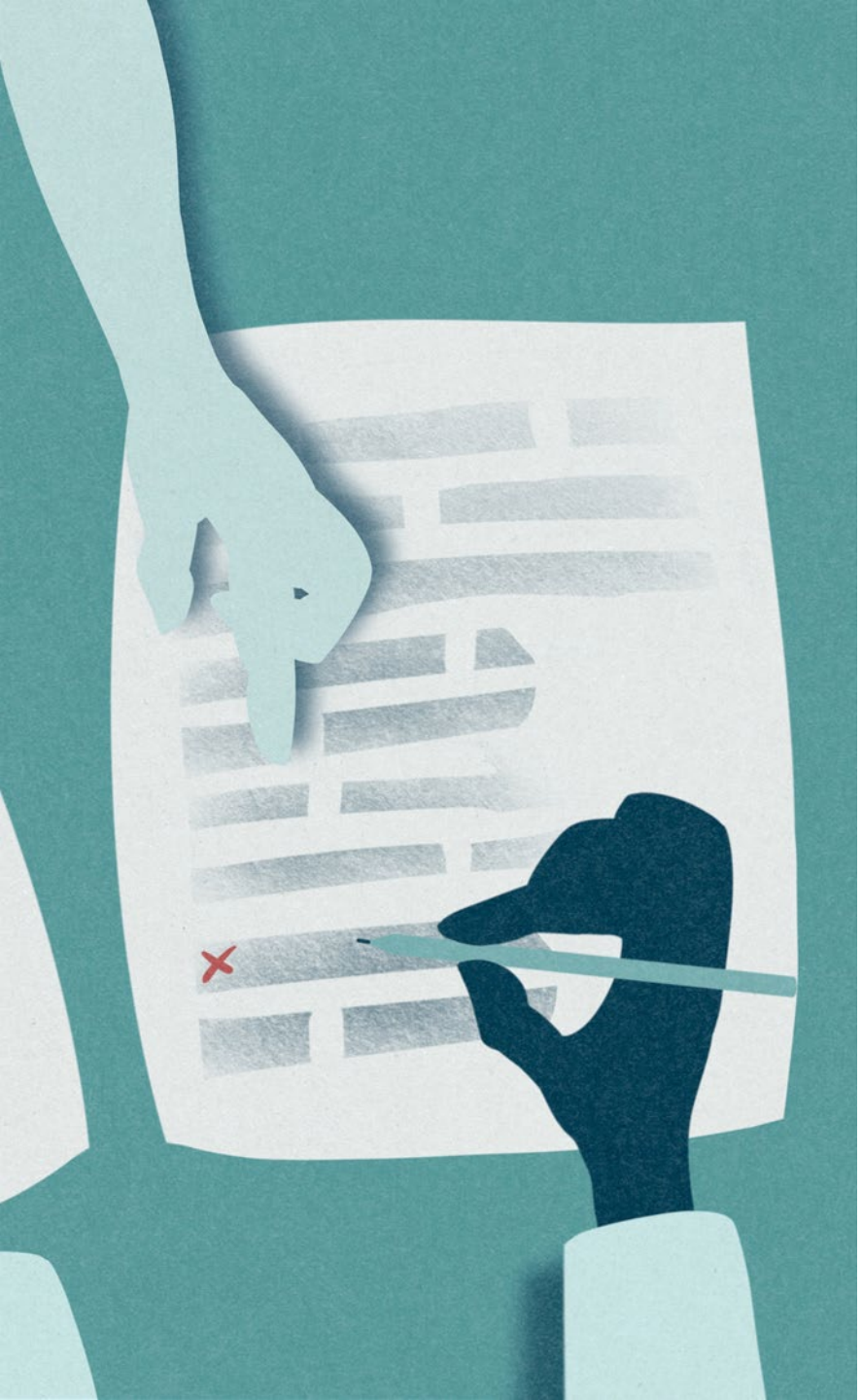
Statistical Product First Initiative

Encouraging a cyclical evaluation of the intended purposes and uses of our statistical products, driving innovation and continual improvement within those products and spawning new and innovative research questions to drive future scientific research.

Future Enhancements

Statistical Products

Need better processes for reviews and decision-making about the promotion or demotion of statistical products to/from core product status.



Without Conflict of Interest or Conflict of Commitment

Already Established

There are well established laws and ethical guidelines regarding financial and personal conflicts of interest in contracting, awards, and hiring.

Future Enhancements

Internal Training

Improve awareness about broader issues relating to conflicts of interest—particularly those regarding conflicts of commitment.

Develop a process for raising and addressing potential conflicts that could impede adherence to the tenets.

Transparent

Already Established

Presentation of current research at scientific and professional association meetings and other public forums

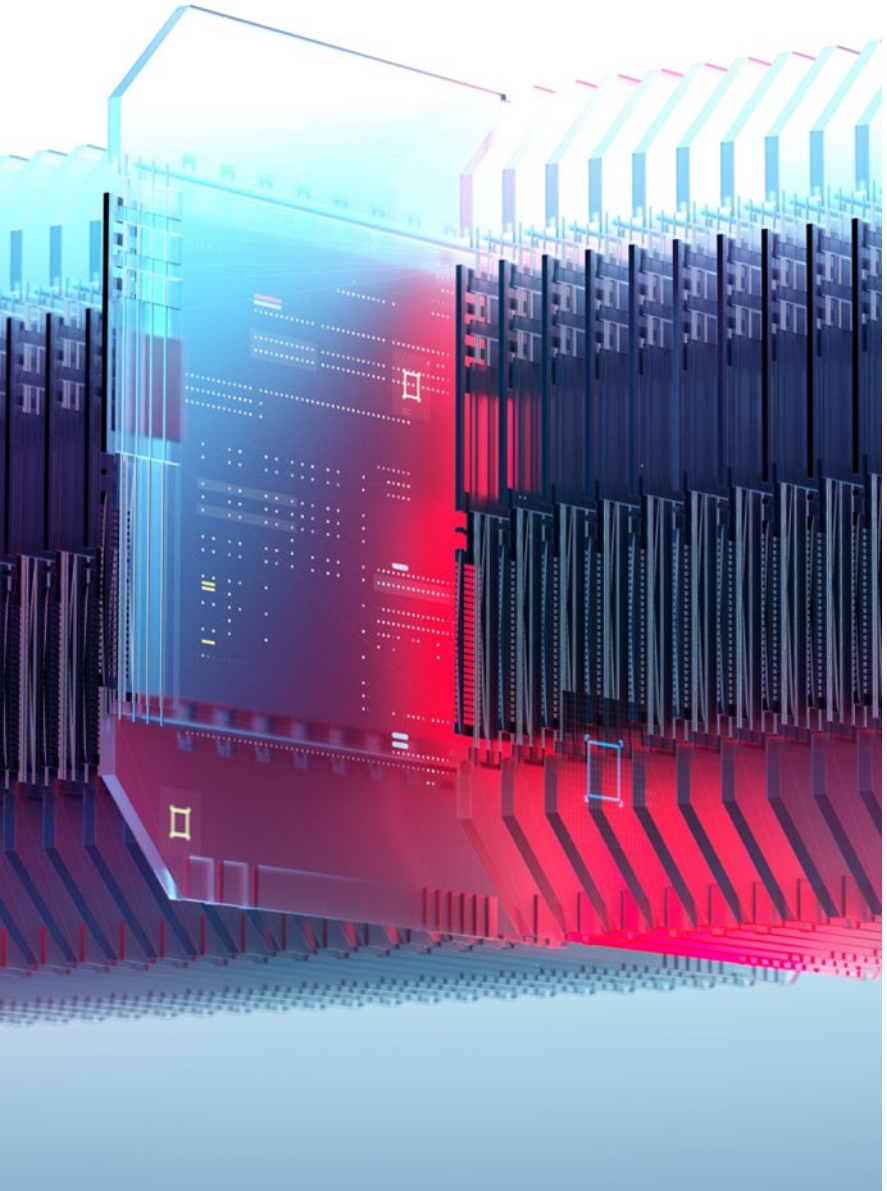
Concept of Operations for Open Census Initiative

Future Enhancements

Open Census implementation—better documentation, traceability, and attribution for components of intramural and sponsored extramural research

Stakeholder engagement—expansive, sustained, and coordinated mechanisms for engaging stakeholder groups regarding scientific activities and research

Record schedules—update to ensure all intramural and sponsored extramural research is properly archived and preserved



Reproducible

Already Established

Policy on Research Transparency and Public Access (DS027)—affirms the Bureau's commitment to ensuring intramural and sponsored extramural research is:

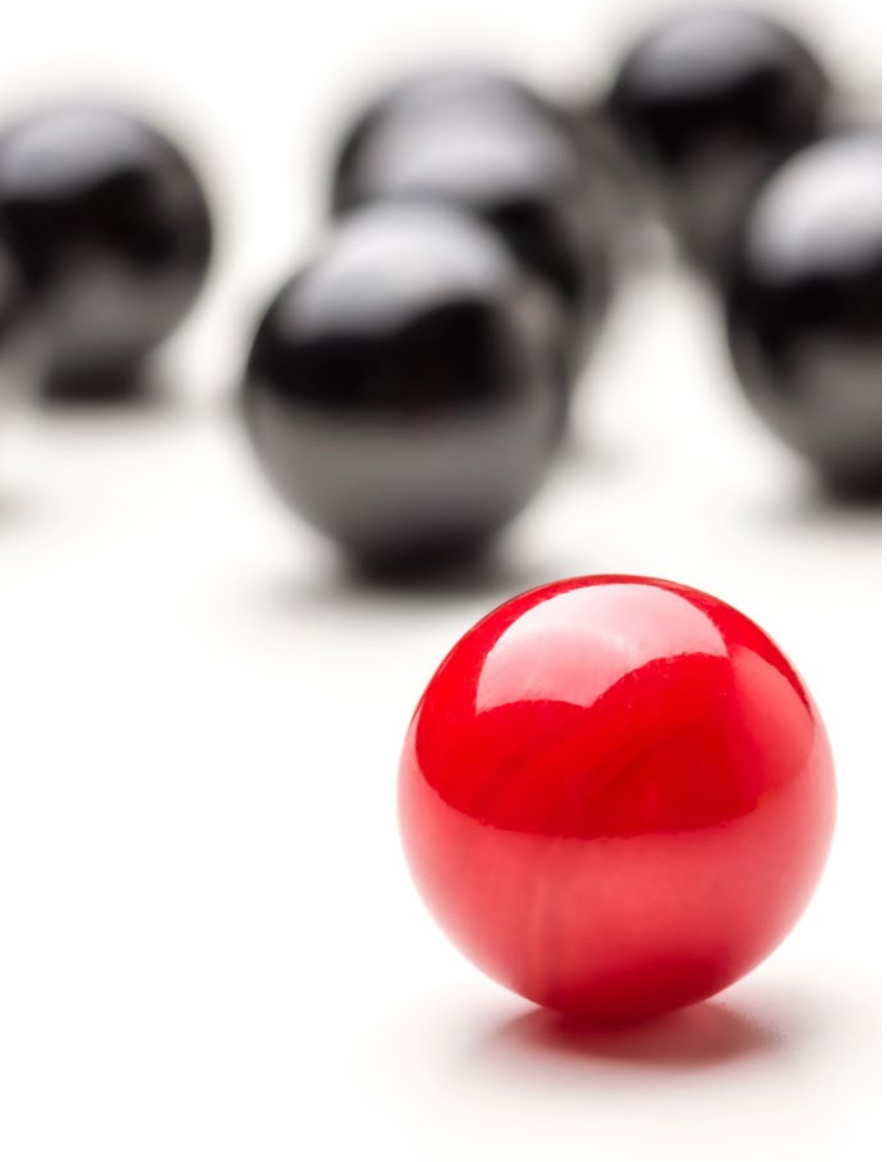
- Relevant
- Credible
- Transparent
- Objective
- Open

Future Enhancements

Open Census Initiative— Implementation of the Initiative's Open Code and Open Data repositories, to include deliberate archiving, curation, and vintaging of data, will be a cornerstone of the Census Bureau's ability to support reproducibility and replicability of research.

Training—Software development and coding best practices as well as better versioning and curation of production code and data files will facilitate internal replication of research and statistical processing

FSRDC Program—Providing access to confidential data for peer-reviewed journals to replicate researcher analyses



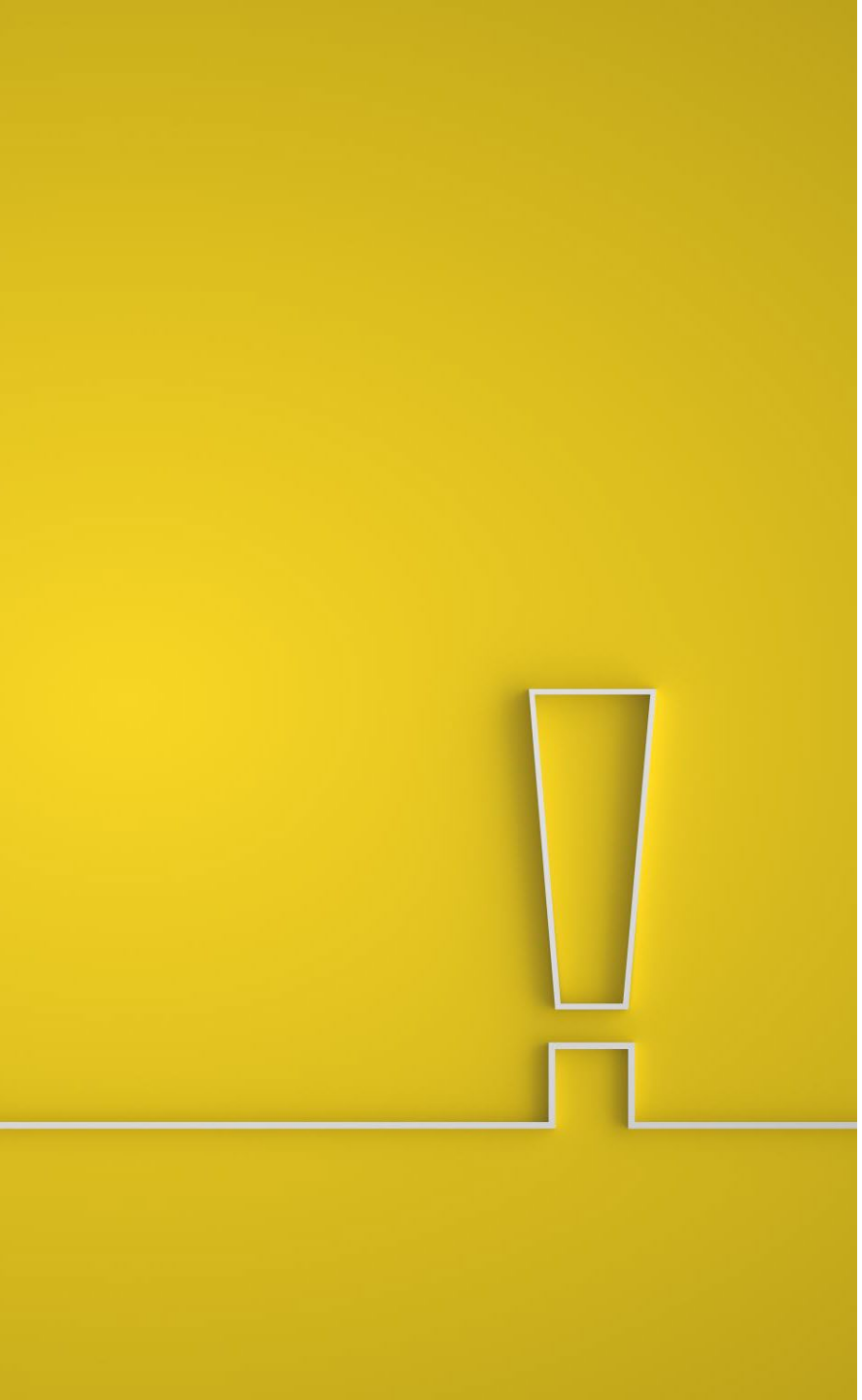
Accepting of Negative Results as Positive Outcomes

Already Established

Simultaneous investment in multiple research paths with the understanding that not all research will pan out.

Future Enhancements

Open Census Initiative—While the broader research community is generally not supportive of publishing negative findings, the Census Bureau must cultivate the expectation among agency and affiliated researchers that negative results *should be* published in the Census Bureau's Open Science research repositories.



Communicative of Error and Uncertainty

Already Established

Sample surveys publish margins of error

Future Enhancements

Statistical Quality Framework Initiative—need frameworks for defining quality and quantifying and communicating errors beyond traditional margin of error (e.g., operational error, disclosure avoidance error)

Methodology and Standards Council—through its working groups must develop standards for quantifying uncertainty for non-traditionally sourced products and better standards for communicating uncertainty and inferences

Data User Tools—Need improved mechanisms for assisting users to incorporate measures of error and uncertainty into their analyses

Skeptical of Findings and Assumptions

Already Established

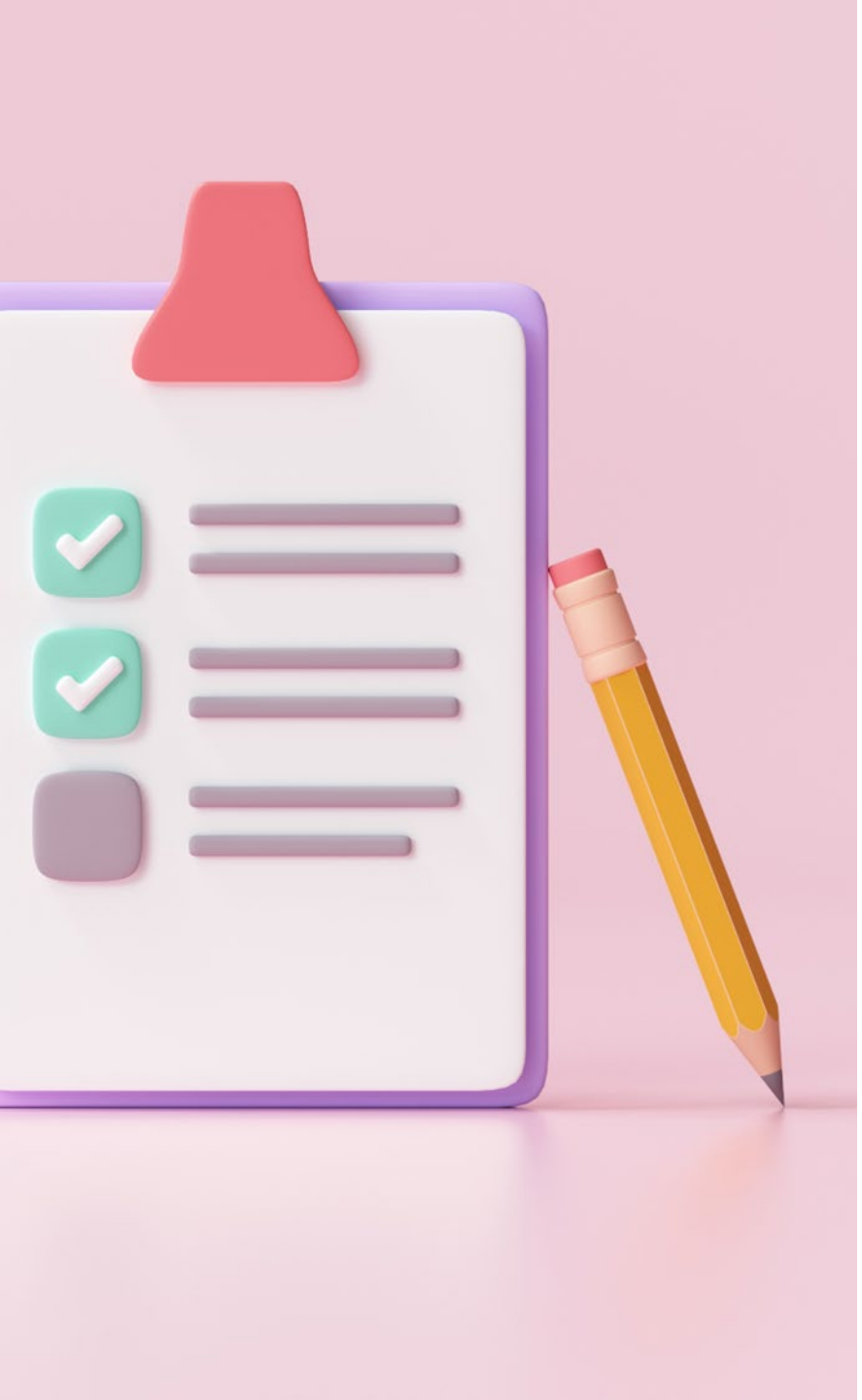
Methodology and Standards Council (interdisciplinary group that reviews early-stage research and experimental products)

Waiver request process for dissemination of statistical products when findings and assumptions are incongruent or standards are not being met

Future Enhancements

External vetting and assessment of methodologies and research—avoiding methodological “group think” requires sustained engagement of Census Bureau researchers with external scientific communities

Training—develop and promote bureau-wide training on the role of science in Bureau activities, stressing the importance of scientific rigor and the responsible use of science



Unbiased Peer Review

Already Established

Statistical Quality Standards—Standard E-3 establishes internal review requirements for all Census Bureau information products.

Encourage external dissemination and submission to peer-reviewed journals.

Future Enhancements

Internal Peer Reviews—Need improved standards for ensuring unbiased reviews beyond the supervisory chain, especially when reviewers are closely involved with the research or methodologies being reviewed.

Engagement with professional associations—Need to expand and sustain researcher engagement with professional associations for junior and mid-level researchers.

Broader dissemination of research throughout the research lifecycle

Open code/replication packages

Questions

United States Census Bureau